Suman poddar

Part 1

- **Write how Python is an interpreted programming language.**
  Python is an interpreted language because its code is executed line by line by the Python interpreter without the need for prior compilation.
- **Who created Python and when was it first released?**
  Python was created by Guido van Rossum and was first released in 1991**.**
- **What is the latest version of Python?**
  The latest stable version of Python (as of 2025) is Python 3.13.
- **What was the motivation behind creating Python and what problems was it designed to solve?**
  Python was created to provide a simple, readable, and efficient programming language that emphasizes code readability and reduces program complexity compared to languages like C or Java.

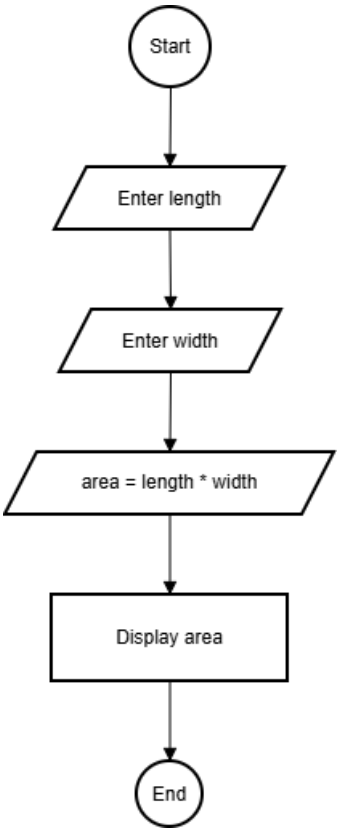
Part 2

1 Write a pseudocode and produce a flowchart to calculate the area of the rectangle.

1.1 Pseudocode

```
START

  PROMPT "Enter the length of the rectangle" → length

  PROMPT "Enter the width of the rectangle"  → width

  COMPUTE area = length * width

  DISPLAY "Area of the rectangle is: " + area

END
```

1.2 FlowChart



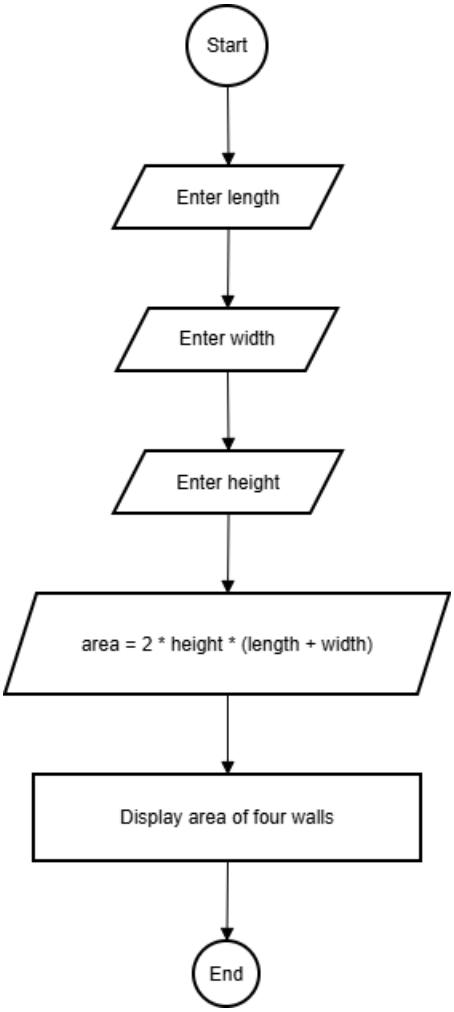
2 Write a pseudocode and produce a flowchart to display the area of four walls.

2.1 Pseudocode

```
START

  PROMPT "Enter the length of the room" length

  PROMPT "Enter the width of the room"   width

  PROMPT "Enter the height of the room"  height

  COMPUTE area = 2 * height * (length + width)

  DISPLAY "Area of four walls is: " + area
```

END

2.2 Flowchart



3 Write a pseudocode and produce a flowchart to calculate the area of the rectangle.

3.1 Pseudocode

START

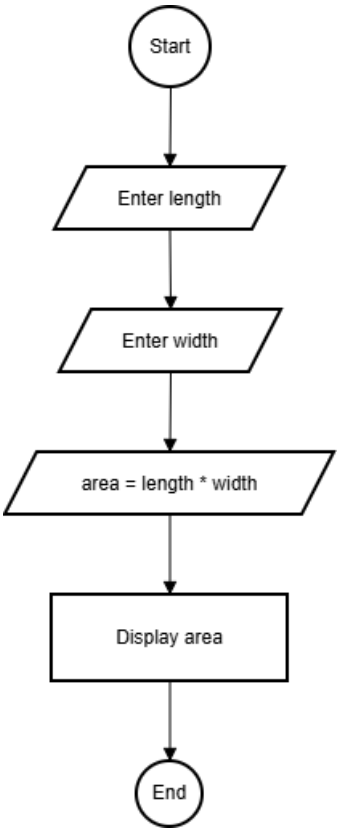
PROMPT "Enter the length of the rectangle" → length

PROMPT "Enter the width of the rectangle" → width

COMPUTE area = length * width

```
  DISPLAY "Area of the rectangle is: " + area

END
```

3.2 Flowchart

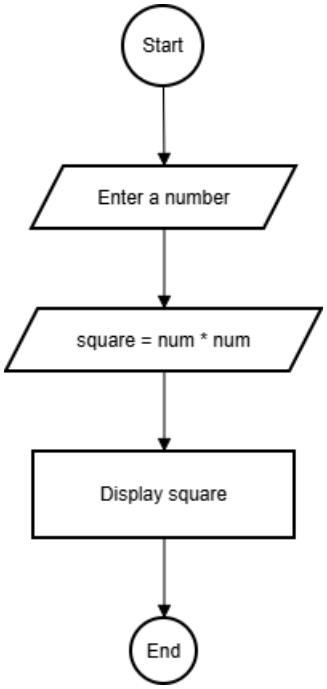


4 Write a pseudocode and produce a flowchart to find the square of a number.

4.1 Pseudocode

```
START

  PROMPT "Enter a number" num

  COMPUTE square = num * num

  DISPLAY "Square of the number is: " + square

END
```

4.2 Flowchart

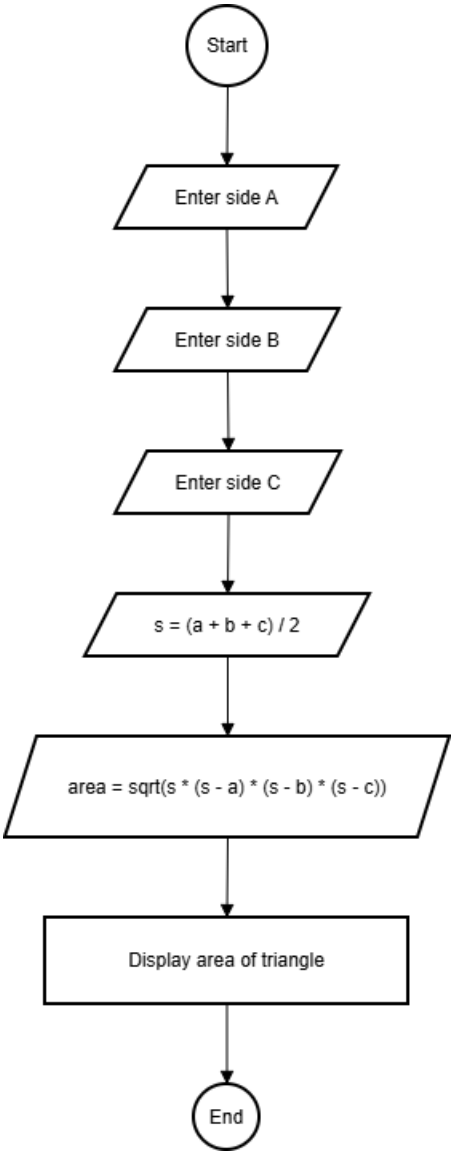


5 Write a pseudocode and produce a flowchart to calculate the area of a triangle when 3 sides of the triangle are given.

5.1 Pseudocode

```
START

  PROMPT "Enter side A" → a

  PROMPT "Enter side B" → b

  PROMPT "Enter side C" → c

  COMPUTE s = (a + b + c) / 2

  COMPUTE area = sqrt( s * (s - a) * (s - b) * (s - c) )

  DISPLAY "Area of the triangle is: " + area

END
```

5.2 Flowchart



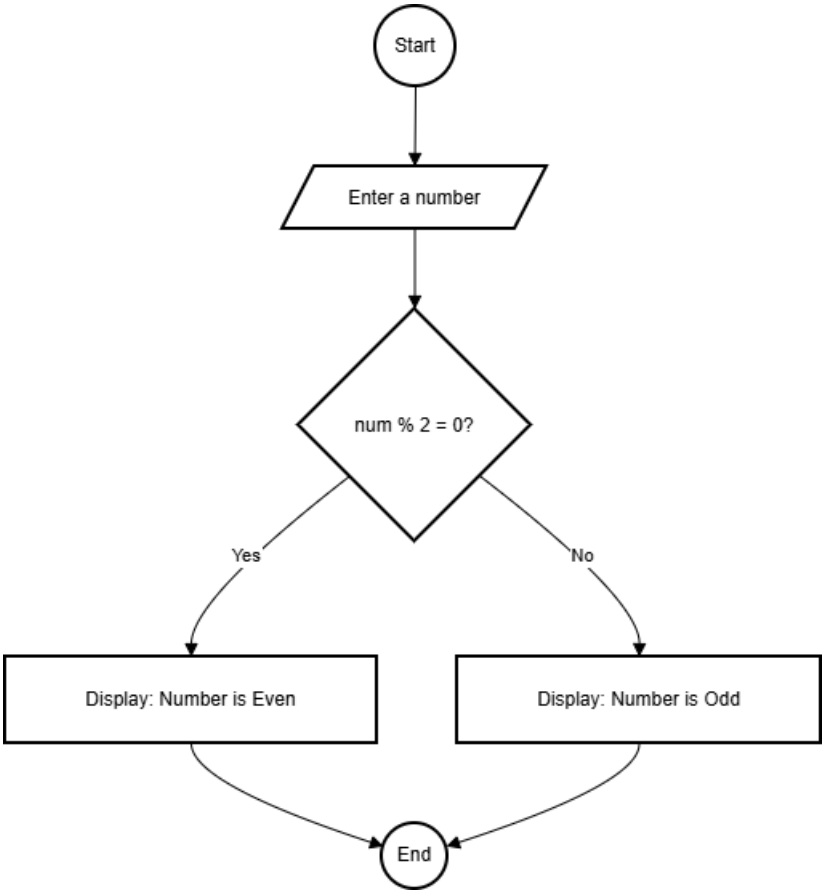
Part 3

6 Write a pseudocode and produce a flowchart to check whether a number is odd or even.

6.1 Pseudocode

```
START

  PROMPT "Enter a number" num

  IF num % 2 = 0 THEN

      DISPLAY "The number is Even"

  ELSE

      DISPLAY "The number is Odd"

  ENDIF

END
```

6.2 Flowchart



7 Suppose you are collecting money for something. You need Rs. 200 in all. You ask your parents, uncles, and aunts as well as grandparents. Different people may give either Rs. 10, Rs. 20, or even Rs. 50. You will collect till the total becomes 200. Write the algorithm.

7.1 Algorithm

1. Start
2. Initialize total = 0

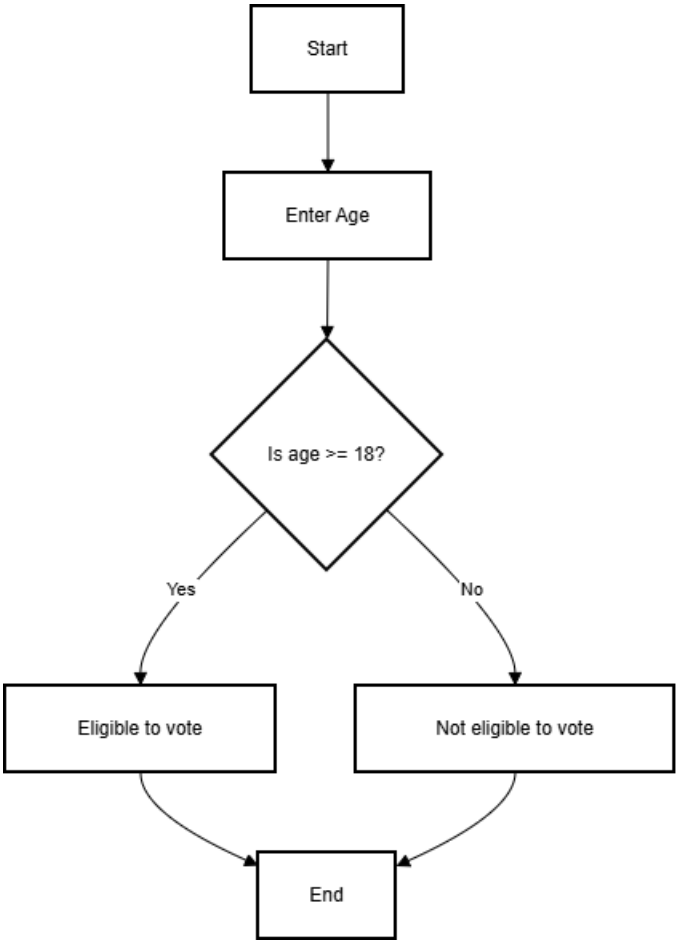
3. Repeat the following steps until total becomes 200 or more:
   a. Ask a family member how much they want to give (Rs. 10, 20, or 50)
   b. Add the given amount to total
4. When the total reaches 200 or more, stop collecting
5. Display the total amount collected
6. End

8 Write a pseudocode/flowchart to identify if a person of a given age is eligible to cast a vote or not.

8.1 Pseudocode

```
START

   PRINT "Enter age:"

   INPUT age

   IF age >= 18 THEN

      PRINT "Eligible to vote"

   ELSE

      PRINT "Not eligible to vote"

   END IF

END
```

8.2 Flowchart

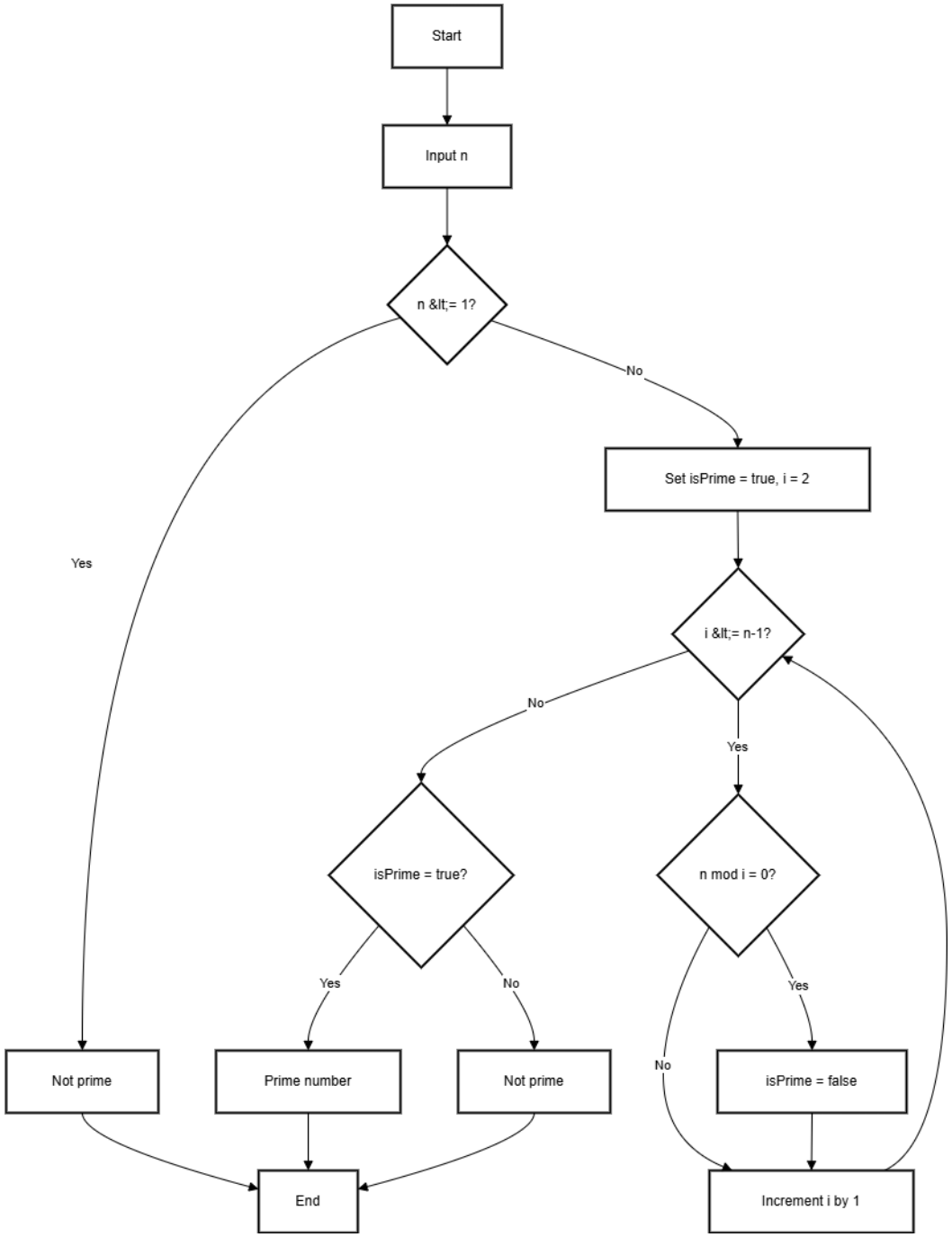


9 Write a pseudocode/flowchart to identify if a number is prime or not.

9.1 Pseudocode

```
START
  INPUT n
  IF n <= 1 THEN
    PRINT "Not prime"
  ELSE
    isPrime true
    FOR i ← 2 TO n-1 DO
      IF n MOD i = 0 THEN
        isPrime false
      END IF
    END FOR
    IF isPrime = true THEN
      PRINT "Prime number"
    ELSE
      PRINT "Not prime"
    END IF
  END IF
END
```

9.2 Flowchart



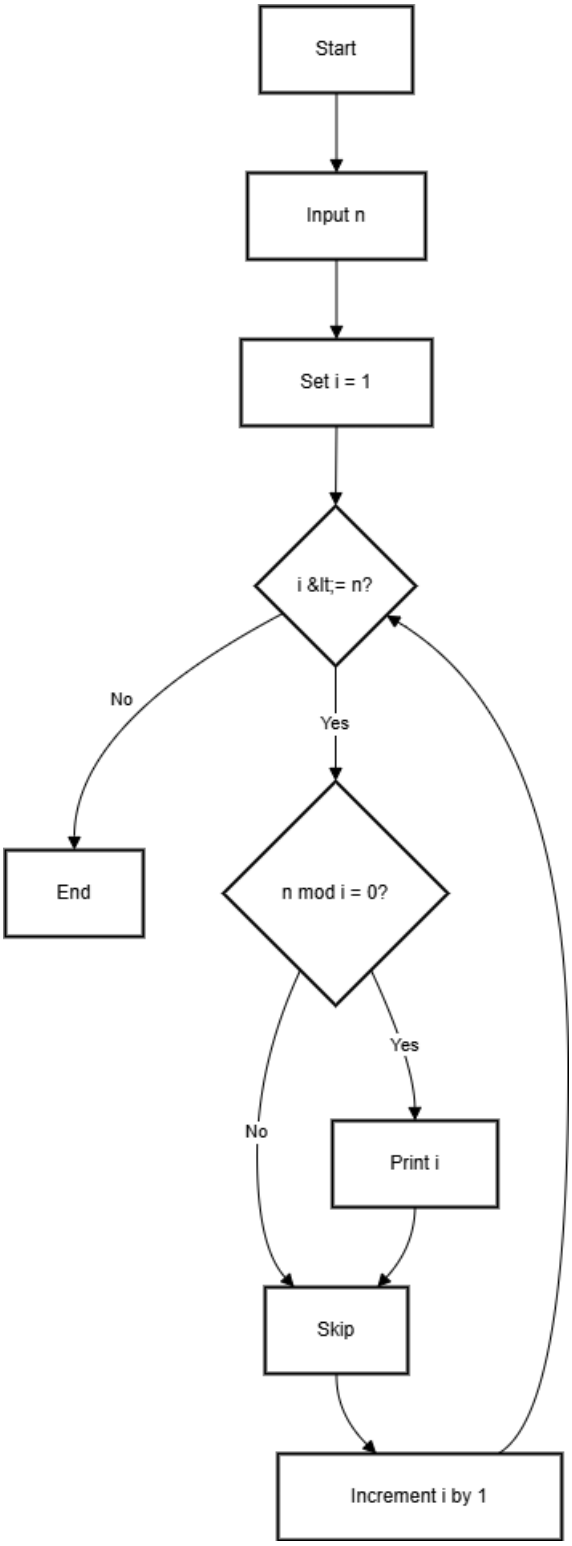
Part 4

10 Write a pseudocode/flowchart to print factors of a given number.

10.1 Pseudocode

```
START

   INPUT n

   FOR i  1 TO n DO

      IF n MOD i = 0 THEN

         PRINT i

      END IF

   END FOR

END
```

10.2 Flowchart

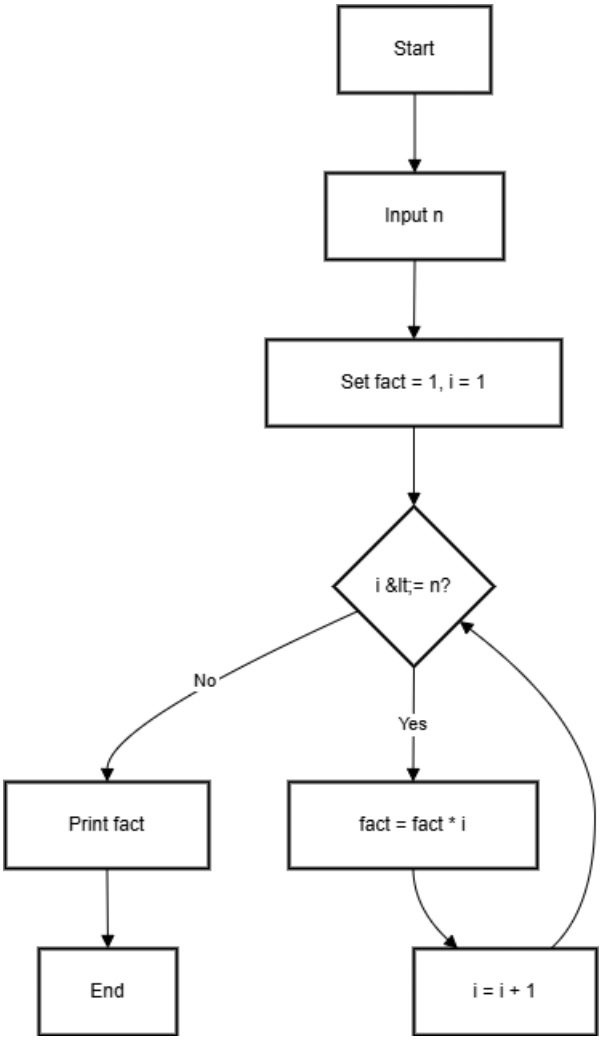


11 Create an algorithm and a flowchart that will output the factorial of a given number.

11.1 Algorithm

1. Start
2. Input the number **n**
3. Set **fact = 1**
4. Set **i = 1**
5. Repeat steps 6 and 7 while **i ≤ n**
6. Multiply **fact = fact × i**
7. Increment **i = i + 1**
8. Display **fact**
9. End

11.2 Flowchart

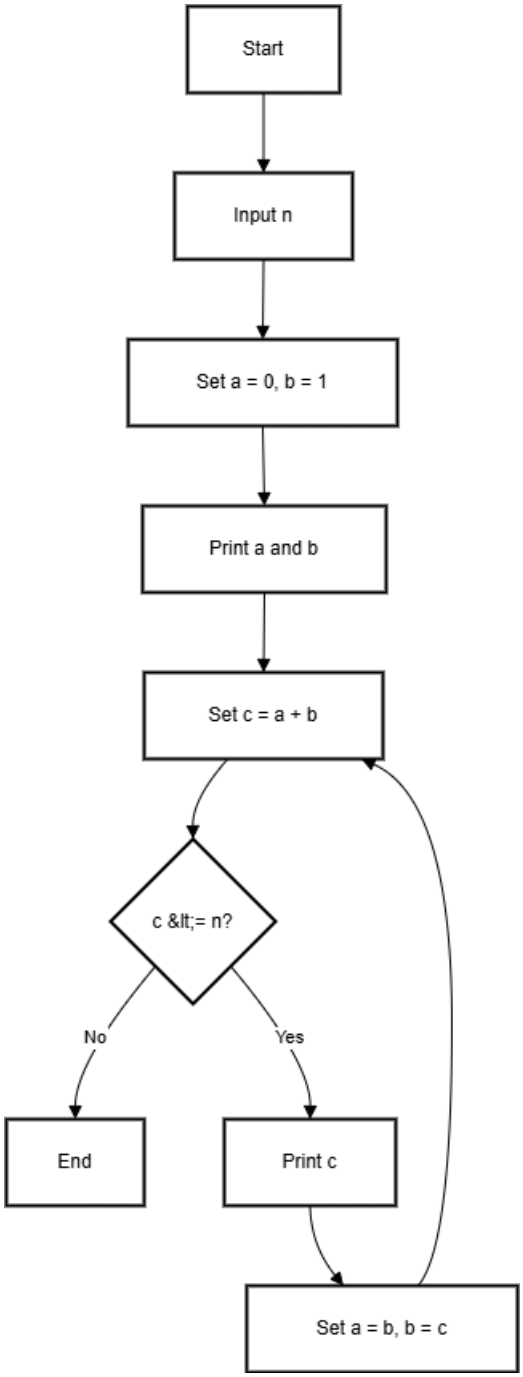


12 Create an algorithm and a flowchart that will output the Fibonacci series up to a given number.

12.1 Algorithm

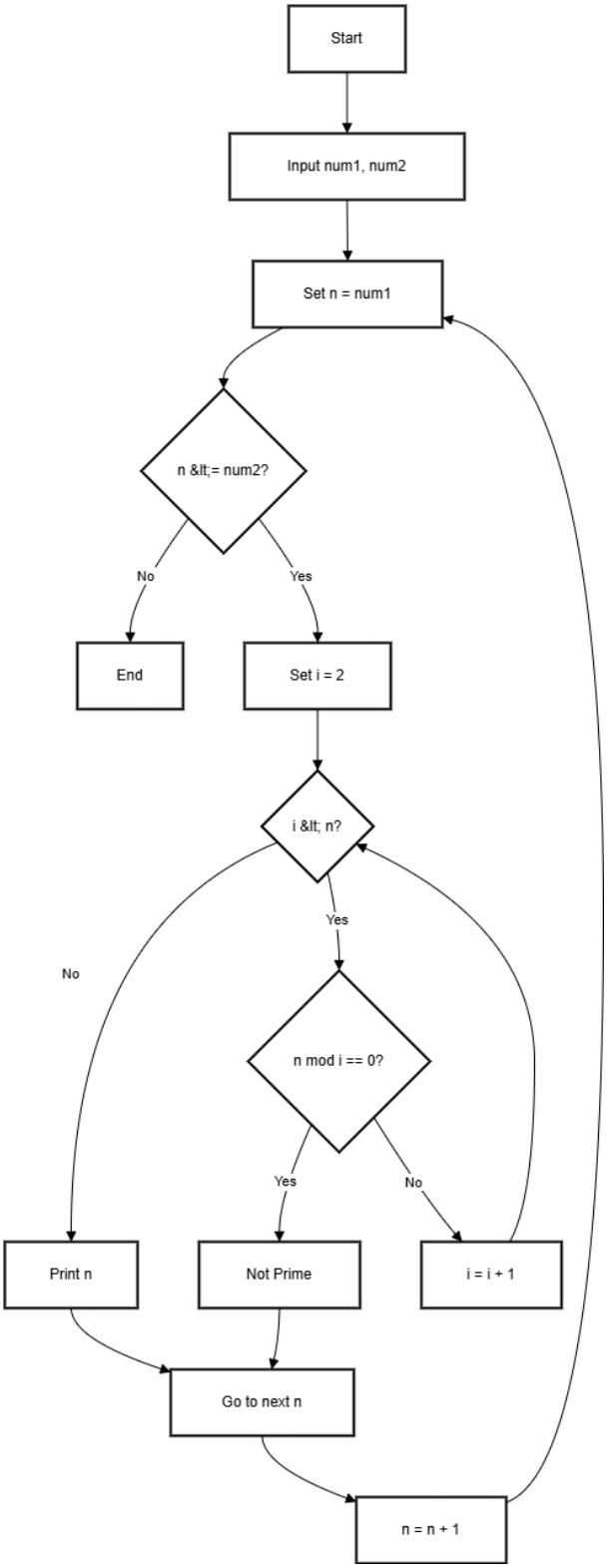
1. Start
2. Input **n**
3. Set **a = 0**, **b = 1**
4. Print **a** and **b**
5. Repeat steps 6–8 while **c ≤ n**
6. Set **c = a + b**
7. Print **c**
8. Set **a = b**, **b = c**
9. End

12.2 Flowchart



13 Create an algorithm and a flowchart that will output all the prime numbers between 2 numbers.

13.1 Algorithm

1. Start
2. Input **num1** and **num2**
3. For each number **n** from **num1** to **num2**, repeat steps 4–8
4. If **n < 2**, skip to next number
5. Set **isPrime = true**
6. For **i = 2** to **n − 1**, do
7. If **n mod i = 0**, set **isPrime = false**
8. If **isPrime = true**, print **n**
9. End

13.2 Flowchart